

Macguire McDuff

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EDUCATION

University of Florida

Bachelor of Science in Computer Science and Engineering

Gainseville, FL

Expected May 2025

- GPA: 3.4; Dean's List (multiple semesters) and SEC Athletic Honor Roll (all semesters)
- Key Classes: Operations Systems, Enterprise Software Engineering, Machine Learning, and Database Systems

NCAA Swimmer

- 5 Time NCAA Champion, 12 Time NCAA All-American, and Current NCAA Record Holder

EXPERIENCE

USA National Team Member

United States Olympic & Paralympic Committee

April 2024 – July 2024

Colorado Springs, CO

- Competed in the final of the Men's 100 Meter Freestyle at Lucas Oil Stadium in front of 22,000+ fan's.
- Trained for 32 hours weekly for Olympic trials, including pool time and participation in wide range of fitness and endurance building activities including weightlifting and running
- Undertook media training and appeared on multiple cable and network stations (including ESPN and NBC)

Software Engineer Intern

Advantus Corp

May 2023 – July 2023

Jacksonville, FL

- Collaborated within a small, agile team of developers to optimize and maintain proprietary systems, utilizing PHP to deliver scalable solutions that drive business integration and innovation.
- Demonstrated expertise in version control by mastering the ins and outs of Git, ensuring seamless collaboration, code quality, and deployment processes.
- Performed maintenance and enhancement tasks on Oracle database platforms, ensuring high availability, data integrity, and efficient compliance with regulatory standards.

Lifeguard

Dupont Employees Recreation Club Incorporated

Summers 2018/2019

Parkersburg, WV

PROJECTS

Allirater | TypeScript, Go, HTML, CSS, Angular

[Github](#)

- Led a team of four to design and develop AlliRater, a platform enabling UF students to rate and access meal reviews for university dining halls, enhancing transparency and user engagement.
- Built a REST API using Angular and Go, ensuring seamless user interactions and efficient data management.
- Implemented Agile methodologies, including sprints and retrospectives, to drive continuous improvement, successfully managing hundreds of unique ratings for reliable, real-time decision-making.

ML Student Performance Predictor | Python, Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow

[Github](#)

- Led a team of four to develop machine learning models for predicting student exam performance, utilizing Python-based tools like scikit-learn, TensorFlow, and pandas for data processing, modeling, and evaluation.
- Conducted regression and classification tasks to predict numerical scores and pass/fail outcomes, leveraging metrics such as MSE, R2, accuracy, and F1 score to assess model performance, achieving top results of 97.03% accuracy.
- Explored 20 key student performance factors (e.g., hours studied, attendance, parental involvement) and implemented data preprocessing techniques on 6,607 records to ensure robust insights for educational support.

Data Analysis of US Housing Market Trends | Oracle SQL, JavaScript, HTML, CSS

[Github](#)

- Collaborated in a team of four to design and implement a robust database system using Oracle SQL, storing and querying over 600,000 tuples of housing and population data to analyze US housing market trends and nationwide economic indicators from 2000 to 2019
- Developed an intuitive, responsive front-end interface with JavaScript, HTML, and CSS, enabling users to execute complex queries and visualize insights into housing, population, and economic factors at both local/national levels.
- Engineered an ERD diagrams to capture housing and economic relationships to support statistical analysis.

SKILLS

Languages: C++, Java, JavaScript, HTML/CSS, Python, SQL, Go, PHP, ADA

Frameworks: Angular, Node.js, React.js, scikit-learn, TensorFlow

Developer Tools: Linux, Git, AWS, Docker, Oracle SQL